Scaling to a Regional Housing Readiness Dashboard

2a. Scaling & technical workflow

The primary data trap: schema fragmentation across jurisdictions. The pilot data already shows it — `Zoning_Mstr` mixes `CITY -RM1-45` with `COUNTY -Rural Residential, Small Agriculture-1`: different naming schemes, different density logic, different vintages, in one field. Expanding to Missoula County and unincorporated Frenchtown, Lolo, and Bonner multiplies that: each jurisdiction's zoning, future land use, and assessor attributes were built for different purposes, and unincorporated areas may have no adopted land-use plan at all. Naively stacking layers produces a dashboard that silently compares incomparable things — the most dangerous failure mode, because it *looks* finished.

The workflow to address it:

1. **Canonical schema first.** Define the region-wide analysis schema (the pilot's derived fields: developable acres, plan-enabled capacity, opportunity class) before acquiring anything.
2. **Per-jurisdiction crosswalks as data, not code.** Each jurisdiction gets a version-controlled crosswalk table (source class ⟶ canonical class ⟶ assumed density, with a source citation per row). Crosswalks are reviewed with planning staff from each jurisdiction — the local-knowledge check no script can replace.
3. **One automated pipeline.** The pilot's R pipeline generalizes: raw sources are never edited; every transformation is scripted; adding a jurisdiction means adding a config block and a crosswalk, not new code.
4. **Validation gates between stages.** Automated checks at each step — row counts reconcile, CRS uniform, geometries valid, no unmapped classes, capacity totals within plausibility bounds per jurisdiction — so errors surface at ingest, not at the client meeting.

2b. Delegation & quality control

Role	Owns	Why it fits
Junior Associate	Data acquisition and management; crosswalk drafting per jurisdiction; analysis runs; source documentation	Judgment-building work with clear review points
GIS Intern	Geometry QA (validity, topology, CRS); metadata inventory; map production from templates; dashboard layer prep	Well-specified, checkable tasks that build real skills
Me (strategic lead)	Schema and assumption sign-off; crosswalk reviews with jurisdiction staff; client communication; final QC	Keeps my time on decisions, not production

Weekly cadence: Monday stand-up, mid-week working session, Friday review of outputs against the task list.

Quality control before anything reaches the client:

- **Automated:** the pipeline's validation gates run on every rebuild; no hand-edited files anywhere in the chain.
- **Peer review:** every crosswalk and analysis script reviewed by a second person; intern's geometry QA spot-checked against county records.
- **Ground-truthing:** for each jurisdiction, a random sample of ~25 parcels checked by hand against assessor records and aerial imagery.
- **Sensitivity disclosure:** headline numbers published with their sensitivity to the key thresholds (density assumptions, softness cutoff), so no single assumption can quietly drive the story.
- **Sign-off checklist:** sources cited, numbers reconciled to pipeline output, cartography standards met — before any deliverable ships.

2c. Budget & schedule ($50,000 fixed, ~16 weeks)

Weeks	Task (deliverable)	Lead	Budget
1–2	1. Scoping, canonical schema, data-sharing letters & named contacts per jurisdiction (schema memo)	Senior	$4,500
2–6	2. Data acquisition & crosswalks, 5+ jurisdictions (crosswalk workbook)	Junior Assoc.	$9,500
5–9	3. Pipeline generalization, capacity analysis, hot spots (validated regional dataset)	Senior + Junior	$11,000
8–13	4. Web dashboard build, open-stack: Quarto/Observable + hosted tiles (beta dashboard)	Senior + Intern	$10,500
11–14	5. Documentation, methods report, QC & ground-truthing (methods report)	All	$5,500
14–16	6. Client reviews, revisions, training & handoff (training + reproducible handoff)	Senior	$4,000
—	Contingency (10%)	—	$5,000
16 weeks	**Total**		**$50,000**

Budgeted at blended rates — Senior Associate $150/hr (~130 hrs), Junior Associate $95/hr (~190 hrs), GIS Intern $55/hr (~135 hrs) — ≈455 staff hours ($45,000) plus $5,000 contingency (10%), sized to the real risk: other jurisdictions' data. Invoicing is tied to the milestone deliverable named in each task, not to hours.

The budget's shape reflects the risk: data normalization (Tasks 1–3, $25,000) is half the project, because the dashboard is only as credible as the crosswalks under it. The schedule's critical path is jurisdiction response time — outside our control — so Task 1 secures data-sharing letters and a named contact per jurisdiction *before* analysis dates are promised. Scope boundary: the fixed fee covers the named jurisdictions and one dashboard revision cycle; additions are priced at the stated rates by change order. All required data is public and free (Montana State Library cadastral, city/county GIS, GTFS, ACS); commercial data, travel, and post-handoff hosting are excluded, with any records-request fees absorbed by contingency. The dashboard stays on an open stack — no per-seat licenses to inherit — and Task 6 ends with the client able to re-run the pipeline when next year's assessor roll lands.